advantage of holding bonds decreases. By contrast, if the recession leads to a higher (perceived) probability of default of LIBOR panel banks, for example, due to their borrowers experiencing trouble or their collateral dropping in value, LIBOR–ASW should decrease. If both effects occur simultaneously, they could net out or one of the two sides, counter-cyclical or cyclical, could prevail, mainly depending on the extent to which banks are affected by the recession.

- If a recession is accompanied by stress in the financial sector, it is probable that haircuts will be increased and that at the same time the cost of financing them will go up. Both effects support increasing swap spreads in a recession and hence reinforce the counter-cyclical impact of the credit risk of bond issuers.

Altogether, in a recession, swap spreads tend to increase due to the worsening of the credit quality of the bond issuer and the tightening haircut schedules (and financing conditions). The impact of a recession on swap spreads via BIS rules and repo–LIBOR spreads is less clear and can work both ways. As a result, while there is a general tendency for swap spreads to be counter-cyclical (increasing during a recession), their behavior during a specific economic cycle depends on the extent to which the recession affects the banking sector. For example, if a recession starts with a banking crisis, it is possible that the worsening credit quality of banks will cause a decrease in LIBOR–ASW, which might be followed by an increase as the recession continues and ends up also affecting the sovereign credit quality (and leads to larger haircuts).

Cutting through Table 12.1 by columns, we note that SOFR–ASW are only affected by counter-cyclical factors (and the non-cyclical variable specialness): both the credit quality of sovereign issuers and haircuts support higher SOFR–ASW in a recession. OIS–ASW and in particular LIBOR–ASW are subject to additional influencing factors, which can include cyclical ones. Hence, one may expect the counter-cyclical behavior of SOFR–ASW to be more pronounced and more predictable than that of LIBOR–ASW.

Moreover, since the financial crisis the credit quality of sovereign issuers has been subject to more scrutiny, as reflected in higher and more volatile CDS levels even for the US and Germany. In terms of Table 12.1, this means that the counter-cyclical factor "credit risk of bond" has gained increasing influence on swap spreads. Consequently, the counter-cyclical behavior of all types of swap spreads has generally increased. And as the transition to SOFR limits the factors influencing swap spreads to counter-cyclical ones, it seems reasonable to anticipate an even clearer and stabler counter-cyclical picture of SOFR–ASW in future, compared to the large variety of patterns displayed by LIBOR–ASW some years ago.

As there has been no full macro economic cycle since the transition, we cannot provide comprehensive empirical evidence for the theoretical forecast of counter-cyclical SOFR–ASW based on our model. At least the semi-cycle shown in Figure 12.4 is in line with it: during the recent economic downturn both the SOFR–ASW of US Treasuries and the CDS level of the US have increased. Unlike in a usual recession, this has occurred against the background of rising interest rates and hence been cyclical. But the argument above has been based on a recession in general and holds true independent of the direction of interest rates, i.e. whether increasing SOFR–ASW during an economic downturn are accompanied by falling (usual) or rising (unusual) yields.

DRIVING FORCES OF SWAP SPREADS CAPTURED IN OUR MODEL

Many analysts use a different approach to assess swap spreads and regress them empirically versus a number of explanatory variables. A typical result could be that swap spreads tend to be higher in times of larger bond issuance, leading them to include Treasury supply as an explanatory variable in a (multiple) regression model for swap spreads.

Most cyclical and counter-cyclical variables are linked to and some of them even caused by the credit quality of the sovereign issuer. This is quite obvious in case of the amount of Treasury supply: the worse the situation of state finances, the larger the debt issuance. Likewise, the higher the unemployment, the lower the tax revenue and the higher the social benefit payments, thus the worse the government's credit situation. In other words, cyclical and counter-cyclical variables like the unemployment rate and government bond supply are both serving as proxies for any credit risk that is reflected in the prices of US Treasury bonds.[10]

Consequently, we claim that our financial model for asset swap spreads not only offers deeper causal insights than a superficial correlation exercise, but also incorporates most of the cyclical and counter-cyclical variables correlated to swap spreads (in particular the amount of bond issuance) via the adjusted CDS level (see Figure 10.1). We see therefore no reason to expand our model with cyclical or counter-cyclical variables, which can reasonably be linked to the credit risk of the government.

[10]We'd like to thank Antti Ilmanen for making this point.

DRIVING FORCES OF SWAP SPREADS NOT CAPTURED IN OUR MODEL

On the other hand, there are a number of driving forces, which are not covered by the conceptual approach of our model. This section provides a (probably incomplete) list of these elements, some of which can and should be added to our model when necessary. In fact, the remaining chapters of the "swap block" will largely be occupied with expanding the asset swap spread concept founded here.

Bonds Trading Away from Par

The presentation of the concept above tacitly assumed par-coupon bonds. The situation becomes a bit more complex in the case of bonds trading at a discount to par, as we'll see in Chapter 17. But fair swap spreads for off-coupon bonds can be calculated by first calculating a fair swap spread for a par-coupon bond and then determining the fair yield spread between the par-coupon bond and the off-coupon bond.

Also, we ignored some of the real-world issues in the repo market. For example, current par-coupon bonds are likely to trade away from par at some point, in which case they will be re-priced in the repo market, as we discussed in Chapter 11. If the swap spread for the bond remains constant, then the NPV of the swap is very likely to move in the opposite direction and by a similar magnitude. Provided the swap is being margined (as almost surely will be the case), then the change in margin for the swap is very likely to offset the change in margin for the bond. For example, if cash were acceptable as margin for the bond position and the swap position, then we could shift the cash from our swap margin account to our repo margin account, and vice versa. However, the swap spread is likely to move over time as well, in which case there will be times when our repo margin and swap margin don't offset.

Insurance Properties of Swap Spreads

So far in our discussion, we've addressed the notion of random and time-varying asset swap spreads by focusing on the expectation of funding and credit spreads over the lives of the bonds. In theory, we might also care about the covariance between these swap spreads and the marginal utility of a typical investor, which is an impact on swap spreads not captured in our model and not treated in the remaining chapters.

When considering this issue, it's useful to draw an analogy to popular types of insurance contracts, such as fire insurance. For most homeowners, the expected return on a fire insurance policy is negative, in that they are required to pay more in insurance premiums than the payouts they expect to receive from the policy, given the likelihood of a fire. Nevertheless, most homeowners are happy to buy fire insurance because the payouts from a policy come at precisely the time that the homeowner most needs money to rebuild a home destroyed by fire. In other words, the covariance between the payout and the homeowner's marginal utility is relatively high.

As discussed above, swap spreads are subject to a number of cyclical and counter-cyclical forces and hence correlated with the marginal utility of the typical investor. For example, in a recession, when banks are at a higher risk of experiencing stress, swap spreads (at least SOFR–ASW) typically increase. In order to assess their covariance to marginal utility, the volatility of swap spreads and marginal utility also needs to be considered: As volatility is usually higher in times of a recession, we'd expect the insurance component of swap spreads to be greater during times of recession than it would be when the economy was expanding.

As a result, a "fair" swap spread might also reflect the extent to which the spread is correlated with factors investors may want to hedge. For example, a bank may have a preference to reverse asset swap bonds, because the (SOFR–)ASW is likely to widen in bad times. Of course, the range of considerations here will be quite complex. For example, if the fortunes of the bank's swap counterparty are likely to be correlated with the macroeconomy, this factor may affect the bank's willingness to enter into the swap agreement as well. But the point underlying both issues is that the performance of the swap may be correlated with things that the bank may wish to insure. And just as insurance policies tend to be priced so that the providers of insurance can expect to make a profit, so should we expect swaps to be priced so that the providers of insurance can expect to make a profit.

While the potential impact of the insurance properties of swap spreads on their pricing cannot be denied, it cannot be observed or modeled: if a subset of market participants finds that reverse asset swapping bonds helps insure against macroeconomic risk, how are the other market participants even to know of this, let alone incorporate it into their own analysis? Like the shadow costs discussed in Chapter 18, this is therefore a driving force which necessarily remains outside any model framework.

Global Influences via the CCBS

By construction, our model looks at the situation in one currency only and we can repeat our exercise for each currency. But due to the existence of CCBS linking all currencies, the swap spreads and their modeling in a given currency are subject to influences from all other swap spread markets.

This means that we cannot simply multiply our model and have one model for the US, a separate one for the eurozone, etc., but that we need to consider at the same time the mutual links via the CCBS between all of them. This is an essential, relatively complex but also worthwhile task, which is the subject of Chapter 16.

Repo Haircuts and Other Individual Factors

While we have included repo haircuts in the qualitative discussions of term structure and cyclicality, we have not included them in our quantitative formula. The reason for this decision is the fact that both the financing costs of repo haircuts and the applicable haircut schedule can vary significantly between individual market participants. Conceptually, one can think of the formula above as representing "the" swap spread, using only input variables which tend to be rather similar for most typical market participants,[11] which then needs to be adjusted with individually different parameters, such as the haircut, in order to obtain "your" swap spread.

These impacts of individual factors, including haircuts, on swap spreads will be the subject of Chapter 18, helping the reader to adjust the general pricing formula according to his specific circumstances.

Altogether, the model presented in this chapter is only the foundation for expansions and modifications in the following chapters. The increasing complexity reflects the complex interdependencies established by the CCBS and the tricky features of the CDS, among others. Unfortunately, already the core model of this chapter contains an unobservable variable (the cost of equity) and while most of the subsequent additions are observable (like ICBS and CCBS), the adjustments of the CDS will involve some more assumptions based on only a few market observations during defaults. Fortunately, we will encounter on our journey through this web of complex relationships and partly unobservable variables some hard and clear boundaries, specifically the arbitrage inequality in Chapter 16.

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[11] As also the CDS quotes can vary depending on the credit standing of the counterpart, it is hard to draw a scientifically precise line between individual and non-individual factors.

CHAPTER 13

Credit Default Swaps

INTRODUCTION

Credit default swaps (CDS) have the advantage of providing observable market prices for credit exposure. We have exploited this advantage by using them as input into our swap spread model in Chapter 12. Unfortunately though, they are not a pure measure of credit risk, but in the case of sovereigns also involve an FX component and in case of physical delivery a delivery option. The first task of this chapter is therefore to assess the contribution of these elements to the observed CDS prices, allowing us to extract the pure credit information from the CDS quotes and to use them as input variable "adjusted CDS" in our swap spread model.

In principle, the data needed to adjust for the FX component and the delivery option are observable ex ante (unlike the CoE, for instance). However, as a consequence of the scarcity of CDS quotes in the domestic currency and of relevant sovereign defaults, there are only a few observations. The adjustments are therefore based on a few data points only and can be improved by adding more observations in future. The best estimation for the 'adjusted CDS' input variable currently possible given the rare observations seems to be:

- The 'pure' credit risk amounts to 45% of the CDS quote observed in the market in case of a sovereign CDS with cash settlement.
- The 'pure' credit risk amounts to 39% of the CDS quote observed in the market in case of a sovereign CDS with physical delivery.

In relation to Chapter 12, this chapter thus provides (an imperfect estimation for) the input variable 'adjusted CDS' for the swap spread model. In the following chapters, this model will be expanded with the global links established by the CCBS. One result will be the ability to express any bond in any currency via a combination of asset&basis swaps as a swap versus USD SOFR. As this is conceptually very similar to a CDS, which also expresses any bond as a spread versus USD SOFR, it is natural to compare the two. CDS will therefore reenter the stage in Chapter 16 discussing the RV relationships between asset, basis and credit default swaps. Specifically, by comparing the USD ASW with the CDS (i.e. by comparing credit risk reflected in the bond market *relative* to credit risk reflected in the CDS market), one can detect relative value opportunities, exploiting the *different assessment* of the same credit risk in bonds and in CDS, while being hedged against the absolute level of credit risk.

Apart from being a driving force of swap spreads, CDS are also an interesting asset class of their own:

- CDS markets exhibit intrinsic relationships, which are well suited to be analyzed and traded via PCA.
- By subtracting (adjusted) CDS from bond yields, one obtains a proxy for risk-free government yield curves. This is the basis for several analytical insights and relatively new RV trading strategies.

These two points are the subject of the second part of this chapter, which can be skipped by readers focused on understanding swap spreads.

STRUCTURE OF A CDS

A CDS contains information about a bond issuer's credit risk, but not in a clean form. Rather, a CDS presents the information about credit risk in connection with other elements, which arise from its structure and legal specifications. Here, we briefly discuss the specifications of the CDS as far as they are relevant for the potential distortion of the credit information. We shall find that the credit information in a CDS is given:

- together with information about the delivery option (DO) in case of CDS with physical delivery;
- potentially in a different currency than the bond covered by the CDS.

Then, we address these two issues separately and investigate how these two factors can be priced and thus how the clean information about credit risk can be extracted from the CDS quotes observed in the market.

A CDS on a bond issuer exchanges floating payments (usually quarterly[1]) of the premium (which we shall also denote with the term "CDS") for the right to deliver a bond of the issuer for payment of its par value in case of a default of the issuer. We shall refer to paying the premium and enjoying the right to deliver the bond for payment of principal in case of a default as "buying" or "being long" the CDS.

[1]Since CDS are over-the-counter products, any terms can be agreed on. We focus on the most common terms.

The usual specifications of a CDS result in the following features, which overlay the credit information in a CDS with other elements:

- There are two types of CDS depending on the form of compensation in case of a default. In cash-settled CDS, the buyer of the CDS receives a payment to compensate for the loss of holding a bond of an issuer in default. There are several ways to determine the amount of the loss and hence of the compensation, such as an auction or a poll among dealers. By contrast, the buyer of a CDS with physical delivery has the right to deliver *any* bond (as defined in the contract) of the defaulted issuer to the seller in exchange for receiving the par value. Since he can select the bond he delivers, CDS with physical delivery contain a delivery option, unlike cash-settled CDS. For sovereign CDS, physical delivery seems to be more common and unless otherwise stated we assume physical delivery.

  For corporate CDS, on the other hand, there seems to be a general tendency to move away from physical delivery: with CDS becoming popular and their outstanding volume increasing relative to the size of outstanding bonds of the defaulting issuer, physical settlement has an increasingly distorting impact on bond prices, since it artificially increases demand for the bonds of defaulting issuers, just for them to be delivered into CDS contracts. Actually, in the case of more CDS contracts (net) than bonds being outstanding, the need to deliver a bond in order to receive payment from the CDS contract should theoretically result in the bonds of a defaulting issuer trading at par. This problem of physical delivery has supported the transition to cash settlement for corporate CDS – though it has its own set of problems. For sovereign issuers, the amount of bonds is much larger relative to the net position of CDS contracts, which may explain why the transition to cash settlement is not as significant as for corporate issuers.
- CDS contracts with physical delivery usually do not specify which bond needs to be delivered in the event of default but rather allow any of the issuer's bonds to be delivered into the CDS contract. This results in a delivery option (DO), similar to a bond futures contract, though in this case the buyer of CDS is long and the seller of CDS is short the DO. If we own a bond and a CDS on its issuer, in the event of default we can deliver our bond in exchange for our principal. But we also could sell our bond, buy another bond of the same issuer, and then deliver that bond into the CDS. If the other bond is cheaper than the one we originally held, we realize a profit.
- In most cases, the CDS terminates when an issuer defaults, even if the stated end date of the CDS is still in the future. For example, if we hold a 10-year (10Y) bond and hedge it with a 10Y CDS, and if the bond issuer defaults after five years, then we can deliver the bond into the CDS in

exchange for the bond principal at the time of default rather than waiting until the maturity of the bond. In contrast, cash flows of other instruments traded in conjunction with the bond, such as asset swaps and basis swaps, would not end in the case of default by the issuer. In our example, if we hold a 10Y *asset-swapped* bond and hedge it with a 10Y CDS, and if the bond issuer defaults after five years, then we receive the principal from the CDS counterparty after five years, but we'll no longer receive the bond cash flows that we had matched against our open asset swap position. Thus, we would need to take our principal and reinvest it at the prevailing interest rate after five years in order to fulfill the remaining asset swap contract. Obviously, this introduces an interest rate risk in case of a default.

- Unlike corporate CDS, sovereign CDS usually[2] pay the principal of a defaulted bond *in USD*. For example, the typical Japanese CDS gives the buyer the right to deliver in case of a default any JGB and to receive the notional value of the CDS, which is trading in USD. This agreement therefore amounts to receiving the par value of the JGB in USD at the USD–JPY exchange rate defined when entering the CDS, hence pre default. We refer to the currency in which the principal of a defaulting bond is paid to the buyer of a CDS contract as the settlement currency or *denomination* of the CDS. As a consequence, a non-USD-denominated bond and a USD-denominated CDS assess the credit risk *in different currencies*.

  In the likely case of the currency of a defaulting country weakening this can result in overcompensation for the CDS buyer. The holder of a USD-denominated CDS will not only get his principal back but very likely receive the principal in a stronger currency. This results in an additional profit, which should be reflected in the pricing of the CDS. In fact, if CDS denominated in local currencies trade at all, they tend to quote at a tighter premium than USD-denominated CDS on the same issuer.

  Combining this point with the previous one leads to a specific issue when a 10Y *asset&basis swapped* bond in local currency is hedged with a 10Y *USD-denominated* CDS. A default after five years results in payment of the principal in USD at the FX rate pre default, while the asset&basis swap contracts remain unaffected and require service of the cash flows in local currency. In this case, the default results not only in an open interest rate exposure but also in an open FX exposure, since we need to exchange our USD back into local currency in order to serve the payment obligations in the ASW and CCBS. If the local currency weakens in the event of a

[2]There are JPY-denominated CDS contracts on Japan and EUR-denominated CDS contracts on Italy trading, but liquidity is rather poor. The only exceptions are CDS contracts on the US, which are usually denominated in EUR.

default, we can obtain an additional profit, since through the CCBS and CDS we are over-hedged against the weakening currency.

- Other specifications with a potential impact on the pricing of a CDS include the treatment of coupons, margin and collateral requirements, consideration of the time value for the settlement process, and the potential difference between the frequency of coupon payments and CDS premium payments.

Trying to strip out the pure credit information from the CDS quotes observed in the market, we need at least to obtain a fair value for the DO and the impact of having the CDS and the bond denominated in different currencies. The next two sections address the issues involved in this task.

Delivery Option

Pricing the DO requires two things, a model and historical data. For the first ingredient one could replicate the DO model for bond futures contracts from Chapter 7 for the delivery situation of a CDS. As discussed in Chapter 7, this model requires the price spread volatility of deliverables as an input variable, and its results will largely depend on this input.

While the issues regarding modeling have been solved in Chapter 7, we need to estimate a realistic price spread volatility for deliverables in case of a default and face the basic problem that there are little to no precedents of *relevant* defaults. While the yield spread volatility between cheapest-to-deliver (CTD) candidates in the current Bund futures contract can be estimated by the yield spread volatility between CTD candidates in previous contracts (see Figure 7.4), which historical precedents could we use to get an idea about the price spread volatility between Bunds in the event of a German default?

Using the price spread volatilities between the bonds of defaulting Latin American countries as inputs, our model returns an average DO value of 6 dollars per 100 dollar par CDS contract *in the event of default*. This might mean that about 6% of the CDS quote observed in the market could be due to the DO. However, the significant standard deviation of the DO in the case of Latin American defaults suggests that defaults and the DO value they produce are hardly comparable, even within the Latin American sovereign universe. And using the price spread volatility between a handful of Argentinean or Ecuadorian bonds as an estimate for the price spread volatility between German issues in case of a default is obviously even more problematic.[3]

[3]The data situation is better in the US credit universe; but, again, would the default of a corporation be a relevant precedent to assess the expected price spread volatility in case of a bank defaulting, which involves completely different considerations (e.g. between senior and subordinated debt)?